Joshua Dinn

DATA SCIENCE & MACHINE LEARNING

joshdinn01@gmail.com | +61 429 556 122 | linkedin.com/in/josh-dinn | github.com/dinnyyy | dinnyyy.github.io

EDUCATION

Monash University, Melbourne, Australia — Mar 2023 – Oct 2027

Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

EXPERIENCE

Collingwood Football Club, Data & Analytics Intern — Mar 2026 – Jun 2026

- Built a membership churn prediction model in XGBoost on historical transaction and engagement data, engineering the feature pipelines end to end to flag at-risk members.
- Used SHAP analysis to surface the behaviours driving retention, then built an interactive dashboard breaking churn down by segment so the commercial team could act on it directly.
- Set up a synchronisation workflow between the club's web and mobile repositories to keep deployments consistent across teams.

Monash DeepNeuron, Optimised Computing Member — Sep 2025 – Present

- Software team member on HarvestAI, an autonomous strawberry-picking robotic arm. Curating and augmenting a strawberry image dataset, then training an object detection model to locate individual berries in a live camera feed and classify ripeness in real time, with training accelerated on HPC clusters.
- Evaluating on precision, recall, and F1 against a target detection accuracy of 80%, with segmentation planned next so detection holds up when berries are partly hidden by leaves.
- Built an on-device AI memory system for AR glasses: a vector database with HNSW indexing and approximate nearest-neighbour search powering a low-latency RAG pipeline.
- Teaching Assistant for an 8-week C++ and accelerated computing workshop run with NVIDIA, the Pawsey Supercomputing Centre, and NCI. NVIDIA CUDA certified.

Monash eSolutions, AI Fitness Trainer — Jul 2026 – Present

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Helped deliver a university webinar on AI to an audience of 200+ attendees.

Outlier AI, AI Trainer (Mathematics) — Jun 2024 – Dec 2025

- Reviewed and corrected AI-generated mathematical solutions written in LaTeX to improve accuracy and reasoning quality.
- Wrote detailed feedback on computational and logical errors across advanced problem sets.

PROJECTS

OLS vs MLE: Monte Carlo Study of Estimator Performance — Ongoing

- Designing a Monte Carlo study (Python) comparing OLS against correctly specified maximum likelihood estimators when the classical linear model assumptions fail: heteroscedasticity, skewed and fat-tailed errors, and binary or count outcomes.
- Measuring bias, relative MSE, and empirical coverage of nominal 95% confidence intervals across sample sizes from 25 to 2,000, with thousands of replications per design, to show where standard inference stops being trustworthy.
- Deriving and implementing the Poisson regression MLE from scratch, solving the score equations with Newton-Raphson (IRLS) instead of calling a library fitter.

Volatility Forecasting with a GARCH Model — Apr 2025

- Built a volatility forecasting pipeline for the S&P 500 using a GARCH(1,1) model fitted by maximum likelihood on a decade of daily log returns.
- Validated the fit with Ljung-Box and ARCH-LM residual diagnostics, then forecast conditional variance 100 days ahead, automating data processing, evaluation, and reporting in Python.

Algorithmic Trading System for S&P 500 ETF — May 2025

- Built and backtested a trading system using Hidden Markov Models to classify market regimes for SPY.
- Used RSI, ATR, and volatility signals to drive position sizing and trade execution.
- Deployed it on an AWS Linux server with the Alpaca API for automated paper trading.

SKILLS

Languages: Python, R, SQL, C++, MATLAB, JavaScript

Machine Learning: scikit-learn, XGBoost, PyTorch, OpenCV, SHAP, object detection, model validation

Statistics: Maximum likelihood estimation, Monte Carlo simulation, time-series modelling (GARCH, HMM), panel and cross-sectional methods

Data & Tools: pandas, NumPy, statsmodels, Snowflake, OxMetrics, Excel, AWS, Git, Linux